Enea Manzi

Curriculum Vitae

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Education

09/2024 - 07/2026 — **Master's Degree in Cyber Security**, *University of Milan*, (Milan, IT)
Thesis: "Design and Implementation of an API-Agnostic Tool for REST API Security Assurance"
Supervisors: Marco Anisetti, Claudio Agostino Ardagna
Expected Graduation: 07/2026

09/2021 - 10/2024 — **Bachelor's Degree in System and Network Security**, *University of Milan*, (Milan, IT)
Thesis: "Design and Implementation of a Framework for Evaluating Distributed System"
Supervisors: Marco Anisetti, Filippo Berto
Final Grade: 110/110 cum laude

09/2016 - 06/2021 — **High School Diploma in Computer Science**, *IIS Badoni*, (Lecco, IT)
Final Grade: 93/100

Work & Research Experience

12/2025 - 07/2026 — **Curricular Internship for Master's Thesis**, *SESAR Lab, University of Milan*, (Milan, IT)
Secure Service-oriented Architectures Research Lab (SESAR Lab)
- Addressed the semantic blindness of existing REST API testing tools — which operate on HTTP syntax and miss the OWASP API Top 10 vulnerability classes — by designing **APIGuard Assurance**, an **API-agnostic, contract-driven** tool deriving all target knowledge exclusively from the **OpenAPI Specification (OAS)** and a deployment configuration, with no hardcoded references to any system.
- Implemented the assessment engine: a taxonomy of **eight security domains**, a **DAG-based scheduler** resolving test dependencies across a seven-phase pipeline, and **semantic exit codes** for **CI/CD** integration.
- Tackled the open research problem of automatically evaluating security guarantees by designing **specified oracles** derived from domain-specific knowledge, with results traceable to **OWASP** categories and **CWE** codes.
- Validated the tool on **Forgejo 14.0.3** exposed through **Kong 3.9**: 18 active tests across 7 domains produced 9 PASS, 7 FAIL, 2 SKIP and 98 findings, correctly detecting a real discrepancy between the specification and the target's actual behavior, with **byte-identical** results confirmed across two independent runs — empirically validating the determinism required for integration into continuous development.

03/2024 - 09/2024 — **Curricular Internship for Bachelor's Thesis**, *SESAR Lab, University of Milan*, (Milan, IT)
Secure Service-oriented Architectures Research Lab (SESAR Lab)
- Analyzed existing verification and monitoring solutions in distributed environments and contributed to the design and implementation of the Assurance Engine (a **Rust**-based assurance framework for distributed systems deployed on **Kubernetes**)
- Extended the existing framework to enable integration with **Elasticsearch**, used as the backend for data collection and querying.
- Developed and tested **formal** contracts, based on metrics, for the automated verification of **non-functional properties (NFP)** – including the **CIA triad**, **reliability**, **scalability**, and **performance** in distributed and microservices architectures.
- Implemented **REST APIs** to expose the engine's functionalities.

01/2020 - 02/2020 — **WBL: web developer**, *DaTrik Solutions*, (Casargo, IT)
High school work-based-learning
- Work Based Learning (WBL): development of a web application.
- Technologies used: **HTML**, **JavaScript**, **jQuery**

Projects

APIGuard Assurance – An API-Agnostic Tool for REST API Security Assurance *Master's Thesis Project (A.Y. 2025/2026)*
Designed and engineered as a distributable Python package, with a focus on **type safety**, **static analysis**, and **reproducible builds**.
- Built in **Python 3.12** with **Pydantic** models for runtime data validation and a dynamic test-discovery mechanism (`pkgutil`) requiring zero core changes to add new checks.
- Enforced **mypy strict** typing across the codebase (`disallow-untyped-defs`, `no-implicit-optional`) and verified 100% public-API docstring coverage via AST inspection.
- Isolated an **AGPL-licensed** dependency (`sslyze`) behind an optional install extra in `pyproject.toml`, keeping the core package free of copyleft propagation.
- Packaged with **Hatch**, verifying **byte-reproducible builds** (identical SHA-256 wheel hash across runs) and clean-environment installability.
- Integrated external security scanners (**nuclei**, **testssl.sh**, **sslyze**) through a hierarchical connector architecture separating raw data collection from result evaluation.

Event-Driven IoT Microservices Architecture (Kafka + Kong) *Cloud Computing Technologies Course Project (A.Y. 2025/2026)*
Development of a cloud-native Kubernetes infrastructure for IoT ingestion and monitoring, with a particular focus on **non-functional properties** such as scalability, fault tolerance, and edge-to-core security.
- Microservices architecture based on **Apache Kafka** in KRaft mode (ZooKeeper-less) for asynchronous decoupling and reliable message handling (*Zero Data Loss*).
- Integration of **Kong API Gateway** to implement the *Gateway Offloading* pattern, centralizing Authentication (API Key), Rate Limiting, and Load Balancing at the edge.
- Storage layer optimized on **MongoDB** using **Time Series Collections** and hybrid compression (LZ4/Zstd) to maximize storage efficiency and analytical performance.
- Full **Kubernetes** orchestration managed via **Helm**, featuring automatic scalability (HPA) and advanced security (Secrets, TLS/SASL) to meet non-functional requirements.

FTP Client and Server *Computer Networks Course Project (A.Y. 2022-2023)*
Developed a multi-threaded **FTP-compliant** client-server application in **C** for file transfer, fully interoperable with **FileZilla**.
- Architecture: Multi-threaded server based on **Socket Programming** for concurrent session management.
- Data Modes: Native support for both Active (PORT) and Passive (PASV) data transfer modes.
- Protocol Management: Implemented essential *FTP commands* (e.g., USER, PASS, LIST,RETR, STOR, CWD...)
- **Key skills: C**, **socket programming**, **multithreading**, **network protocols**, **FTP**, **Git**.

Exhibition Web App *Web and Mobile Programming Course Project (A.Y. 2021-2022)*
Built a dynamic web application serving as a digital portfolio for displaying multimedia content.
- Developed the backend with *Node.js/Express* and the frontend with *HTML5*, *CSS3*, *JavaScript/jQuery*.
- Integrated *MongoDB* for document-oriented storage and *AJAX* for asynchronous client-server communication (GET, POST, DELETE).
- Managed source code and collaboration using **Git** and **GitHub**.
- **Key skills:** *Node.js*, *Express*, *MongoDB*, *AJAX*, *REST APIs*, *jQuery*, **Git**.

Publications

TODO: exact title Enea Manzi, Marco Anisetti, Claudio A. Ardagna
ITADATA 2026 (submitted)
TODO: draft below, to refine

Languages

Italian	Mother tongue
English	Intermediate *Listening B2 - Reading B2 - Writing B2 - Spoken Production B2 - Spoken Interaction B2*

Skills

Programming Languages	*C*, *Java*, *Python*, *SQL*, Web (*HTML*, *JavaScript/Node.js*), OOP Concepts, low-level programming
Database	*MySQL*, Database Design, Implementation and Adminsitration

Architectures & Software	*Docker*, *Kubernetes*, microservices architectures, REST API, Git, Kafka, Kong API Gateway
Operating Systems	Fresh OS installation, Linux Administration and Shell, Windows
Networking & Security	Network Protocols, ICT Network Security Risk, Secure Coding Practices
Tools	Visual Studio Code, GitHub

Soft Skills

Thinking	analytically, critically, proactively
Problems	address problems critically, solve problems/create solutions
Communication	cooperate with colleagues, work in teams, manage teamwork
Others	demonstrate curiosity, cope with stress, prioritise tasks

Bellano, June 24, 2026

Enea Manzi